Yearbook of Agricultural Statistics 2020

Curated Extract for Embeddings — Chapter 1: Introduction

Agro-Ecological Zones • Land Levels • Soil Classification • Crop Seasons • Crop Calendar • Physiography

This file is a selective, cleaned extract of Chapter 1 ("Introduction") of the source Yearbook, prepared for building a RAG knowledge base. It covers only sections 1.3-1.9: Agro-Ecological Zones (AEZ), Land Levels in Relation to Flooding, Soil Classification, Crop Seasons, the Crop Calendar of Bangladesh, and Physiography. Section 1.1 (general introduction) and Section 1.2 (BBS statistical survey methodology for crop estimation) were deliberately excluded as not relevant to agronomic/crop-recommendation grounding. The other ~600 pages of the source Yearbook (multi-year production/yield time series tables) are also excluded from this file; those are better parsed directly into a structured reference table (e.g. Postgres) than embedded as text. Tabular data (AEZ table, soil table, crop calendar table) has been reconstructed from the original multi-page tables and cross-checked against the source page images for accuracy, including manual verification of rows that fragmented across a page break. Every entry below cites its original page number(s) in the source PDF: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series.

Section 1.3-1.4 — Agro-Ecological Zones (AEZ) of Bangladesh

The land use pattern of the country is influenced by agro ecology, soil physiographic and climatic factors. According to the variations of all these factors and agricultural potential, the total land area has been classified into thirty agro-ecological zones (AEZ), which are grouped into twenty major physiographic units. Each zone below combines the summary table data (physiographic unit, area, land-type distribution with dominant soil texture/pH/organic matter per land type) with the descriptive profile (landscape, dominant soil types, general fertility) and the districts/upazilas where the zone occurs.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Table 1.3.1 (Physiography and AEZ), pp.25-27]

AEZ-1: Old Himalayan Piedmont Plain

Physiographic Unit: Old Himalayan Piedmont Plain

Area (excluding river beds): 990400 acres

Land type distribution: High 58% (Sandy loam texture, pH 4.5-5.5, Low organic matter); Medium High 34% (Loamy texture, pH 4.5-5.5, Low organic matter); Medium Low 1%; Homestead+Water 7%

Description: This distinctive region is developed in an old Teesta alluvial fan extending out from foot of the Himalayas. It has complex relief pattern comprising broad and narrow floodplain ridges and linear depressions. Deep rapidly permeable sandy loams and sandy clay loams are predominant in the region. They are strongly acidic in topsoil and moderately acidic in subsoil as well as richen sand minerals. Seven general soil types occur in the region; of them, Noncalcareous Brown Floodplain soils, Black Terai soils and Noncalcareous Dark Grey Floodplain soils predominant Organic matter contents are relatively higher than other floodplain areas. The natural fertility of the soils, except the coarse textured, is moderate but well sustained.

Location: Most of Panchagarh and Thakurgaon districts and north western part of Dinajpur district

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.25; Section 1.4 description p.28]

AEZ-2: Active Tista Floodplain

Physiographic Unit: Teesta Flood Plain

Area (excluding river beds): 205100 acres

Land type distribution: High 2%; Medium High 72% (Sandy loam texture, pH 6.0-6.6, Low organic matter); Homestead+Water 26%

Description: This region includes the active floodplains of Teesta, Dharla and Dudkumar rivers. It has complex patterns of low, generally smooth ridges, inter-ridge depressions, river channels and cut-off channels. The area has irregular patterns of grey stratified, sands and silts. They are moderately acidic throughout and parent alluvium is rich in minerals. Four general soil types occur in the region; of which, Noncalcareous Alluvium predominates. Organic matter content is low and CEC is medium. Soil fertility level, in general, is low to medium.

Location: Narrow belts within and adjoining the channels of the Teesta, Dharla and Dudkumar rivers in Nilphamari, Rangpur, Lalmonirhat, Kurigram and Gaibandha districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.25; Section 1.4 description p.28]

AEZ-3: Tista Meander Floodplain

Physiographic Unit: Teesta Flood Plain

Area (excluding river beds): 2339595 acres

Land type distribution: High 35% (Loamy texture, pH 5.4-6.5, Low organic matter); Medium High 51% (Loamy texture, pH 6.0-6.5, Low organic matter); Medium Low 4%; Low 1%; Homestead+Water 9%

Description: This region occupies major part of the Teesta floodplain as well as the floodplains of the Atrai, Little Jamuna, Karatoya, Dharla and Dudkumar rivers. Most areas have broad floodplain ridges and almost level basins. There is an overall pattern of olive brown, rapidly permeable, loamy soils on the high floodplain ridges, and grey or dark grey, slowly permeable, heavy silt loam or silty clay-loam soils on the lower land and parent materials rich in minerals. Eight general soil types occur in the region; of which, Noncalcareous Grey Floodplain and Noncalcareous Brown Floodplain soils predominate. They are moderately acidic throughout, low in organic matter content on the higher land, but moderate in the power parts. Fertility level, in general, is low to medium. Soils in general have a good moisture holding capacity.

Location: Most of greater Rangpur, eastern parts of Panchagarh and Dinajpur, northern Bogura and parts of Joypurhat, Naogaon and Rajshahi districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.25; Section 1.4 description p.28]

AEZ-4: Karatoya-Bangali Floodplain

Physiographic Unit: Teesta Flood Plain

Area (excluding river beds): 635555 acres

Land type distribution: High 23% (Silt-loam texture, pH 5.4-5.7, Low organic matter); Medium High 44% (Clayey texture, pH 5.4-5.7, Low organic matter); Medium Low 14%; Low 4%; Very Low 1%; Homestead+Water 14%

Description: This floodplain apparently comprises a mixture of Tista and Brahmaputra sediments. Most areas have smooth, broad, floodplain ridges and almost level basins. The soils are grey silt loams and silty clay loams on ridges and grey or dark grey clays in basins. Five general soil types occur in the region; of which, Noncalcareous Grey Floodplain and Noncalcareous Dark Floodplain soils predominate. They are moderately acidic throughout. Organic matter content is low in ridge soils and moderate in basins. General fertility is medium.

Location: Eastern half of Bogura and most of Sirajganj districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.28]

AEZ-5: Lower Atrai Basin

Physiographic Unit: Lower Atrai Basin

Area (excluding river beds): 210285 acres

Land type distribution: High 2%; Medium High 8%; Medium Low 21% (Clayey texture, pH 4.8-6.0, Medium organic matter); Low 65% (Clayey texture, pH 4.8-6.0, Medium organic matter); Homestead+Water 4%

Description: This region comprises the low-lying area between the Barind Tract and the Ganges River Floodplain. Smooth low lying basin land occupies most of the region. Dark grey, heavy, acidic clays predominate. Seven general soil types occur in the region; but Noncalcareous Dark Grey Floodplain soils cover most of the area. Organic matter and fertility status are moderate.

Location: Naogaon and Natore districts and parts of Rajshahi, Bogura and Sirajgonj districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.29]

AEZ-6: Lower Punarbhaba Floodplain

Physiographic Unit: Lower Punarbhaba Flood Plain

Area (excluding river beds): 31875 acres

Land type distribution: Medium High 10%; Medium Low 60% (Clayey texture, pH 4.5, Medium organic matter); Homestead+Water 30%

Description: This region occupies basins and beels separated by low floodplain ridges. In this area, dark grey, mottled red, very strongly acid, heavy clays occupy both ridge and basin sites. Only one general soil type, Acid Basin Clays, has been identified in the region. Organic matter status is medium to high. General fertility level is medium.

Location: Extreme western part of Naogaon district and extreme northern part of Chapainawabgonj district.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.29]

AEZ-7: Active Brahmaputra-Jamuna Floodplain

Physiographic Unit: Brahmaputra Flood Plain

Area (excluding river beds): 788265 acres

Land type distribution: High 5%; Medium High 37% (Sandy/Silty texture, pH 7.5-7.9, Very low organic matter); Medium Low 20% (Silty texture, pH 7.5-7.9, Low organic matter); Low 8%; Homestead+Water 30%

Description: This region comprises the belt unstable alluvial land along the Brahmaputra-Jamuna rivers where land is constantly being formed and eroded by shifting river channels. It has an irregular relief of broad and narrow ridges and depressions. The area is occupied by sandy and silty alluvium rich in minerals with slightly alkaline in reaction. Six general soil types occupy the area; of which, only Noncalcareous Alluvium predominates. Organic matter status is low and fertility status low to medium.

Location: Eastern parts of Kurigram, Gaibandha, Bogura, Sirajganj and Pabna districts and western parts of Sherpur, Jamalpur, Tangail and Manikganj districts.Also, minor areas in Dhaka, Munshiganj, Narayanganj and Chandpur districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.29]

AEZ-8: Young Brahmaputra -Jamuna Floodplain

Physiographic Unit: Brahmaputra Flood Plain

Area (excluding river beds): 1463850 acres

Land type distribution: High 18% (Loamy texture, pH 5.5-6.8, Low organic matter); Medium High 42% (Loamy texture, pH 5.5-6.8, Low organic matter); Medium Low 19% (Loamy texture, pH 5.5-6.8, Low organic matter); Low 9%; Homestead+Water 12%

Description: This region comprises of the area of Brahmaputra sediments. It has a complex relief of broad and narrow ridges, inter-ridge depressions, partially in filled cut-off channels and basins. This area is occupied by permeable silt loam to silty clay loam soils on the ridges and impermeable clays in the basins which are neutral to slightly acidic in reaction. General soil types include predominantly Grey Floodplain soils. Organic matter content is low in ridges and moderate in basins.

Location: Western parts of Jamalpur, Sherpur and Tangail districts, parts of Manikganj Narayanganj and Gazjipur districts and a belt adjoining the old Brahmaputra channel through Mymensingh, Kishoreganj and Narsingdi districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.29]

AEZ-9: Old Brahmaputra Floodplain

Physiographic Unit: Brahmaputra Flood Plain

Area (excluding river beds): 1786570 acres

Land type distribution: High 28% (Silt-loam texture, pH 5.1-5.6, Low organic matter); Medium High 35% (Loamy texture, pH 5.1-5.6, Medium organic matter); Medium Low 20% (Clayey texture, pH 5.1-5.6, Medium organic matter); Low 7%; Homestead+Water 10%

Description: This region occupies a large area of Brahmaputra sediments before the river shifted to its present Jamuna channel about 200 years ago. The region has broad ridges and basins. Soils of the area are predominantly silt loams to silty clay loams on the ridges and clay in the basins. General soil types predominantly include Dark Grey Floodplain soil. Organic matter content is low on the ridges and moderate in the basins; topsoils are moderately acidic but subsoils neutral in reaction. General fertility level is low.

Location: Large areas in Sherpur, Jamalpur, Tangail, Mymensingh, Netrokona, Kishoreganj districts. Also, small areas east of Dhaka and Gazipur districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.29]

AEZ-10: Active Ganges Floodplain

Physiographic Unit: Ganges River Flood Plain

Area (excluding river beds): 823850 acres

Land type distribution: High 12%; Medium High 33% (Loamy texture, pH 6.9-7.9, Low organic matter); Medium Low 18% (Clayey texture, pH 6.9-7.9, Medium organic matter); Low 4%; Homestead+Water 33%

Description: This region occupies unstable alluvial land within and adjoining Ganges river. It has irregular relief of broad and narrow ridges and depressions. The area has complex mixtures of calcareous sandy, silty and clayey alluvium. The general soil types predominantly include Calcareous Alluvium and Calcareous Brown Floodplain soils which are low in organic matter and mildly alkaline in reaction. General fertility level is medium.

Location: The region along the Ganges and lower Meghna River channels from the Indian border in Chapainawabgonj and Rajshahi districts to the mouth of Meghna estuary in Lakshmipur and Barishal districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.29]

AEZ-11: High Ganges River Floodplain

Physiographic Unit: Ganges River Flood Plain

Area (excluding river beds): 3263025 acres

Land type distribution: High 43% (Silt-loam texture, pH 6.1-7.9, Low organic matter); Medium High 32% (Loamy texture, pH 6.1-7.9, Low organic matter); Medium Low 12% (Clayey texture, pH 6.1-7.9, Medium organic matter); Low 2%; Homestead+Water 11%

Description: This region includes the western part of the Ganges River Floodplain which is predominantly highland and medium highland. Most areas have a complex relief of broad and narrow ridges and inter-ridge depressions, separated by areas with smooth broad ridges and basins. There is an overall pattern of olive-brown silt loams and silty clay loams on the upper parts of floodplain ridges and dark grey, mottled brown, mainly clay soils on ridge sites and in basins. Most ridge soils are calcareous throughout. General soil types predominantly include Calcareous Dark Grey Floodplain soils and Calcareous Brown Floodplain soils. Organic matter content in brown ridge soils is low and higher in dark grey soils. Soils are slightly alkaline in reaction. General fertility level is low.

Location: Chapainawabgonj, Rajshahi, Southern Pabna, Kushtia, Meherpur, Jashore, Chuadanga, Jhenaidah, Magura, and northern parts of Satkhira and Khulna districts together with minor areas in Naogaon and Narail districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.30]

AEZ-12: Low Ganges River Floodplain

Physiographic Unit: Ganges River Flood Plain

Area (excluding river beds): 1968935 acres

Land type distribution: High 13% (Silt-loam texture, pH 6.2-7.7, Medium organic matter); Medium High 29% (Silt-clay texture, pH 6.2-7.7, Medium organic matter); Medium Low 31% (Clayey texture, pH 6.2-7.4, Medium organic matter); Low 14%; Very Low 2%; Homestead+Water 11%

Description: The region comprises the eastern half of the Ganges River Floodplain which is low-lying. The region has a typical meander floodplain landscape of broad ridges and basins. Soils of the region are silt loams and silty clay loams on the ridges and silty clay loams to heavy clays on lower sites. General soil types predominantly include Calcareous Dark Grey and Calcareous Brown Floodplain soils. Organic matter content is low in ridges and moderate in the basins. Soils are calcareous in nature having neutral to slightly alkaline reaction. General fertility level is medium.

Location: Natore, Pabna, Goalanda, Faridpur, Madaripur, Gopalganj and Shariatpur, eastern part of Kushtia, Magura, and Narail, north eastern parts of Khulna and Bagherhat, northern Barishal, and south-western part of Manikganj.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.30]

AEZ-13: Ganges Tidal Floodplain

Physiographic Unit: Ganges Tidal Flood Plain

Area (excluding river beds): 4217100 acres

Land type distribution: High 2%; Medium High 78% (Loamy texture, pH 6.5-7.0, Medium organic matter); Medium Low 2%; Homestead+Water 18%

Description: This region occupies an extensive area of tidal floodplain land in south-west of the country. The greater part of this region has smooth relief. There is a general pattern of grey, slightly calcareous, heavy soils on river banks and grey to dark grey, noncalcareous, heavy silty clays in the extensive basins. Noncalcareous Grey Floodplain soil is the major component of general soil types. Acid Sulphate soil also occupies significant part of the area where it is extremely acidic during dry season. In general, most of the topsoils are acidic and subsoils are neutral to mildly alkaline. Soils of Sundarban area are strongly alkaline. General fertility level is high with medium to high organic matter content.

Location: All or most of Barishal, Jhalakathi, Pirojpur, Patuakhali, Barguna, Bagerhat, Khulna, Satkhira districts including Khulna and Bagerhat Sundarban Reserved Forests.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.30]

AEZ-14: Gopalganj-Khulna Beels

Physiographic Unit: Gopalganj-Khulna Beel

Area (excluding river beds): 555245 acres

Land type distribution: High 3%; Medium High 13% (Clayey texture, pH 5.4, Medium organic matter); Medium Low 41% (Clayey texture, pH 5.4, Medium organic matter); Low 28% (Clayey texture, pH 5.4, Medium organic matter); Very Low 11%; Homestead+Water 4%

Description: The region occupies extensive low-lying areas between the Ganges River Floodplain and the Ganges Tidal Floodplain. Almost level, low-lying basins occupy most of the region with low ridges along rivers and creeks. Soils of the area are grey and dark grey acidic heavy clays overlying peat or muck at 25-100 cm. Soft peat and muck occupy perennially wet basin centres. General soil types include mainly Peat and Noncalcareous Dark Grey Floodplain soils. Organic matter content is medium to high. Fertility level is medium.

Location: A number of separate basin areas in Madaripur, Gopalgonj, Narail, Jashore, Bagerhat and Khulna districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.30]

AEZ-15: Arial Beel

Physiographic Unit: Arial Beel

Area (excluding river beds): 35585 acres

Land type distribution: Medium High 13% (Clayey texture, pH 5.4, Medium organic matter); Low 73% (Clayey texture, pH 5.4, Medium organic matter); Homestead+Water 14%

Description: This region occupies a low-lying basin between the Ganges and Dhaleshwari rivers in the south of former Dhaka district. 11 The soils of this area are dark grey, acidic heavy clays. Noncalcareous Dark Grey Floodplain soils are the major general soil type. Organic matter content generally exceeds two percent in the top and subsoil. Available moisture holding capacity is inherently low. The general fertility level is medium to high.

Location: Munshigonj and Dhaka districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.30]

AEZ-16: Middle Meghna River Floodplain

Physiographic Unit: Meghna River Flood Plain

Area (excluding river beds): 384250 acres

Land type distribution: Medium High 8%; Medium Low 29% (Loamy texture, pH 4.9-5.5, Low organic matter); Low 25% (Loamy texture, pH 4.4-5.5, Low organic matter); Very Low 11%; Homestead+Water 27%

Description: This region occupies abandoned channel of the Brahmaputra River on the border between greater Dhaka and Cumilla districts. This region includes islands-former Brahmaputra chars within the Meghna River as well as adjoining parts of the mainland. Soils of the area are grey, loamy on the ridges and grey to dark grey clayey in the basins. Grey sands to loamy sands with compact silty topsoil, occupying areas of old Brahmaputra char. Dominant general type is Noncalcareous Grey Floodplain soils. Topsoils are strongly acidic and subsoils slightly acidic to slightly alkaline. General fertility level is medium.

Location: The region between the southern part of Sylhet Basin and the confluence of the Meghna river with the Dhaleshwari and Ganges rivers covering parts of several districts, Kishoreganj, Brahmanbaria, Cumilla, Chandpur, Narsingdi and Narayangonj.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.31]

AEZ-17: Lower Meghna River Floodplain

Physiographic Unit: Meghna River Flood Plain

Area (excluding river beds): 224620 acres

Land type distribution: High 14% (Silt-loam texture, pH 5.0-6.0, Medium organic matter); Medium High 28% (Silt-loam texture, pH 5.0-6.0, Medium organic matter); Medium Low 31% (Silt-loam texture, pH 5.5-6.0, Medium organic matter); Homestead+Water 27%

Description: This area occupies transitional area between Middle Meghna River Floodplain and the Young Meghna Estuarine Floodplain. The region has slightly irregular relief with little difference in elevation between the ridges and depressions. Soils of this area are relatively uniform. Silt loams occupy relatively higher areas and silty clay loams the depressions. Noncalcareous Dark Grey Floodplain and Calcareous Grey Floodplain soils are major components of general type. Topsoils are moderately acidic and subsoils neutral in reaction. General fertility level is medium to high with low to medium organic matter status.

Location: Chandpur, Lakshmipur and Noakhali districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.31]

AEZ-18: Young Meghna Estuarine Floodplain

Physiographic Unit: Meghna Estuarine Flood Plain

Area (excluding river beds): 2290420 acres

Land type distribution: Medium High 45% (Silt-loam texture, pH 6.1-6.8, Medium organic matter); Medium Low 7%; Homestead+Water 48%

Description: This region occupies young alluvial land in and adjoining the Meghna estuary. It is almost level with very low ridges and broad depressions. The major soils are grey to olive, deep, calcareous silt loam and silty clay loams and are stratified either throughout or at shallow depth. Calcareous Alluvium and Noncalcareous Grey Floodplain soils are the dominant general type. The soils in the south become saline in dry season. Top soils and subsoils of the area are mildly alkaline. General fertility is medium but low in organic matter.

Location: Chattogram, Feni, Noakhali, Lakshmipur, Bhola, Barishal, Patuakhali and Barguna districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.26; Section 1.4 description p.31]

AEZ-19: Old Meghna Estuarine Floodplain

Physiographic Unit: Meghna Estuarine Flood Plain

Area (excluding river beds): 1912595 acres

Land type distribution: High 2%; Medium High 24% (Silt-loam texture, pH 5.0-6.1, Medium organic matter); Medium Low 33% (Loamy texture, pH 5.5-6.5, Medium organic matter); Low 21% (Clayey texture, pH 5.5-6.5, Medium organic matter); Very Low 3%; Homestead+Water 17%

Description: This region occupies a large area, mainly low-lying between south of the Surma-Kusiyara Floodplain and northern edge of the Young Meghna Estuarine Floodplain. It comprises smooth, almost level, floodplain ridges and shallow basins. Silt loam soils predominate on highlands and silty clay to clay in lowlands. Noncalcareous Dark Grey Floodplain soils are the only general type of the area. Organic matter content of the soils is moderate. Moisture holding capacity is medium. Topsoils are moderately acidic, but subsoils neutral in reaction. General fertility level is medium.

Location: Kishoreganj, Habiganj, Brahmanbaria, Cumilla, Chandpur, Feni, Noakhali, Lakshmipur, Narsingdi, Narayanganj, Dhaka, Shariatpur, Madaripur, Gopalganj and Barishal districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.31]

AEZ-20: Eastern Surma-Kusiyara Floodplain

Physiographic Unit: Surma-Kusiyara Flood plain

Area (excluding river beds): 1142120 acres

Land type distribution: High 5%; Medium High 25% (Silt-loam texture, pH 4.7-6.9, Medium organic matter); Medium Low 20% (Clayey texture, pH 4.7-5.5, Medium organic matter); Low 36% (Clayey texture, pH 4.7-5.5, Medium organic matter); Homestead+Water 14%

Description: This region occupies the relatively higher parts of Surma-Kusiyara Floodplain formed on sediments of the rivers draining into the Meghna catchments area from the hills. The area is mainly smooth broad ridges and basins. The soils are grey, heavy silty clay loams on the ridges and clays in the basins. Noncalcareous Grey Floodplain soils are the only general type. Organic matter content of soils is moderate. The reaction of soils ranges from strongly acidic to neutral.

Location: Sylhet, Moulvibazar, Sunamganj and Habiganj districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.32]

AEZ-21: Sylhet Basin

Physiographic Unit: Surma-Kusiyara Flood plain

Area (excluding river beds): 1130015 acres

Land type distribution: Medium High 4%; Medium Low 19% (Silt-loam texture, pH 4.7-4.9, High organic matter); Low 43% (Clayey texture, pH 4.7-4.9, High organic matter); Very Low 23% (Clayey texture, pH 4.7-4.9, High organic matter); Homestead+Water 11%

Description: The region occupies the lower western side of Surma-Kusiyara Floodplain. The area is mainly smooth broad basins with narrow ridges of higher land along the rivers. Soils of the area are grey clays in the wet basins and silty clay loams and clay loam on the higher parts which dry out seasonally. Noncalcareous Grey Floodplain soils and Acid Basin Clays are the major components of the general soil types. The soils have moderate content of organic matter and soil reaction is mainly acidic. Fertility level is medium to high.

Location: Large parts of Sunamganj, Habiganj, Netrokona, Kishoreganj and Brahmanbaria districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.32]

AEZ-22: Northern and Eastern Piedmont Plains

Physiographic Unit: Northern and Eastern Piedmont Plains

Area (excluding river beds): 997810 acres

Land type distribution: High 33% (Sandy-loam texture, pH 4.5-5.8, Medium organic matter); Medium High 31% (Loamy texture, pH 4.5-5.8, Medium organic matter); Medium Low 16% (Silt-loam texture, pH 4.5-5.8, Medium organic matter); Low 9%; Very Low 1%; Homestead+Water 10%

Description: This is a discontinuous region occurring as a narrow strip of land at the foot of the northern and eastern hills. The area comprises merging alluvial fans which slope gently outward from the foot of the hills into smooth low-lying basin. Grey Piedmont soils and Noncalcareous Grey Floodplain soils are the major general soil types of the area. Soils of the area are loams to clays in texture having slightly acidic to strongly acidic reaction. General fertility level is low to medium.

Location: Sherpur, Netrokona, Sunamganj, Sylhet, Moulvi Bazar, Habiganj, Brahmanbaria and Cumilla districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.32]

AEZ-23: Chattogram Coastal Plain

Physiographic Unit: Chattogram Coastal Plain

Area (excluding river beds): 919230 acres

Land type distribution: High 17% (Silt-loam texture, pH 5.6, Low organic matter); Medium High 43% (Silt-loam texture, pH 5.6, Low organic matter); Medium Low 13%; Homestead+Water 27%

Description: This region occupies the plain land in greater Chattogram district and the eastern part of Feni district. It is a compound unit of piedmont, river, tidal and estuarine floodplain landscapes. Grey silt loams and silty clay loam soils are predominant. Acid Sulphate soils which are potentially extremely acidic occur in mangrove tidal floodplains. Noncalcareous Grey Floodplain soils, Noncalcareous Alluvium and Acid Sulphate soils are the major components of general soil types of the area. General fertility level of soils is medium. Organic matter content is low to moderate.

Location: Feni, Chattogram and Cox's Bazar districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.32]

AEZ-24: St. Martin's Coral Island

Physiographic Unit: St. Martins Coral Island

Area (excluding river beds): 1975 acres

Land type distribution: High 33% (Sandy texture, pH 7.0-7.5, Low organic matter); Medium High 63% (Sandy-loam texture, pH 7.0-7.5, Low organic matter); Medium Low 2%; Homestead+Water 2%

Description: This small but distinctive region occupies the whole of St. Martin's Island in the extreme south of the country. The area has very gently undulating old beach ridges and inter-ridge depressions surrounded by sandy beaches. The soils are developed entirely on old and young coral beach sands. Calcareous Alluvium is the only general soil type of the area. General fertility level is low with poor moisture holding capacity.

Location: St. Martin's Island

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.32]

AEZ-25: Level Barind Tract

Physiographic Unit: Barind Tract

Area (excluding river beds): 1247635 acres

Land type distribution: High 30% (Loamy texture, pH 5.0-5.7, Low organic matter); Medium High 55% (Loamy texture, pH 5.0-5.7, Low organic matter); Medium Low 4%; Low 2%; Homestead+Water 9%

Description: This region is developed over Madhupur Clay. The landscape is almost level and locally irregular along river channels. The predominant soils have a grey silty paddled topsoil with plough pan which either directly overlies grey heavy little weathered Madhupur Clay or merges with the porous silt loam or silty clay loam subsoils having strongly acid clay at greater depth. Shallow Grey Terrace soils and Deep Grey Terrace soils are the major components of general soil types of the area. The soils are low in available moisture holding capacity and slightly acidic to acidic in reaction. Organic matter status is very low and most of the available nutrients are limiting.

Location: Dinajpur, Gaibandha, Joypurhat, Bogura, Naogaon, Sirajganj and Natore districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.33]

AEZ-26: High Barind Tract

Physiographic Unit: Barind Tract

Area (excluding river beds): 395370 acres

Land type distribution: High 93% (Loamy texture, pH 4.8-5.9, Low organic matter); Medium High 1%; Homestead+Water 6%

Description: It includes the western part of Barind Tract where the underlying Madhupur Clay has been uplifted and cut into by deep valleys. The soils include paddled silt loam to silty clay loam in the topsoils and porous silt loam with mottled plastic clay at varying depth. Deep Grey Terrace soils and GreyValley soils are the major components of general soil types of the area. General fertility status is low having low status of organic matter.

Location: Rajshahi, Chapainawabgonj and Naogaon districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.33]

AEZ-27: North Eastern Barind Tract

Physiographic Unit: Barind Tract

Area (excluding river beds): 266625 acres

Land type distribution: High 36% (Loamy texture, pH 4.8-5.9, Low organic matter); Medium High 56% (Loamy texture, pH 4.8-5.9, Low organic matter); Medium Low 1%; Homestead+Water 7%

Description: This region occupies several discontinuous areas on the north-eastern margins of Barind Tract. It stands slightly higher than adjoining floodplain land. The region has silty or loamy topsoil and clay loams to clay subsoils and grades into strongly mottled clay. The Madhupur Clay underlying this region is deeply weathered. Deep Red Brown Terrace soil and Deep Grey Terrace soils are the major components of general soil types of the area. The soils are strongly acidic in reaction. Organic matter of the soils is low. General fertility level is poor.

Location: Dinajpur, Rangpur, Gaibandha, Joypurhat and Bogura districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.33]

AEZ-28: Madhupur Tract

Physiographic Unit: Madhupur Tract

Area (excluding river beds): 1048715 acres

Land type distribution: High 56% (Loamy texture, pH 4.8-5.5, Low organic matter); Medium High 18% (Loamy texture, pH 4.8-5.5, Low organic matter); Medium Low 7%; Low 9%; Homestead+Water 10%

Description: This region of complex relief and soils developed over the Madhupur Clay. The landscape comprises level upland, closely or broadly dissected terrace associated with either shallow or broad, deep valleys. Eleven general soil types exist in the area; of which, Deep Red Brown Terrace, Shallow Red Brown Terrace soils and Acid Basin Clays are the major ones. The soils on the terrace are better drained, friable clay loams to clays overlying friable clay substratum at varying depth. Soils in the valleys are dark grey heavy clays. They are strongly acidic in reaction with low status of organic matter, low moisture holding capacity and low fertility level.

Location: Dhaka, Gazipur, Narsingdi, Narayanganj, Tangail, Mymensingh and Kishoreganj.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.33]

AEZ-29: Northern and Eastern Hills

Physiographic Unit: Northern and Eastern Hills

Area (excluding river beds): 4490150 acres

Land type distribution: High 92% (Loamy texture, pH 4.5-4.9, Low organic matter); Medium High 2%; Medium Low 1%; Homestead+Water 5%

Description: This region includes the country's hill areas. Relief is complex. Hills have been dissected to different degrees over different rocks. In general, slopes are very steep and few low hills have flat summits. The major hill soils are yellow-brown to strong brown permeable friable loamy, very strongly acidic and low in moisture holding capacity. However, soil patterns generally are complex due to local differences in sand, silt and clay contents of the underlying sedimentary rocks and in the amount of erosion that has occurred. BrownHill soils are the predominant general soil type of the area. Organic matter content and general fertility level is low.

Location: Mainly in Khagrachhari, Chattogram Hill Tracts, Bandarban, Chattogram, Cox's Bazar, Habiganj and Moulvi Bazar districts, small areas along the northern border of Sherpur, Mymensingh, Sunamganj and Sylhet districts in central and south eastern Sylhet and in the east of Brahmanbaria, Cumilla and Feni districts.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.33]

AEZ-30: Akhaura Terrace

Physiographic Unit: Akhaura Terrace

Area (excluding river beds): 27920 acres

Land type distribution: High 55% (Loamy texture, pH 5.5-6.5, Low organic matter); Medium High 11% (Loamy texture, pH 5.5-6.5, Low organic matter); Medium Low 10%; Low 15%; Very Low 3%; Homestead+Water 6%

Description: This small region lies on the eastern border of Brahmanbaria and southwest corner of Habiganj districts. In appearance, the region resembles Madhupur Tract with level upland dissected by mainly deep broad valleys. The main soils on the upland have strong brown clay which grades into red mottled clay substratum. The valley soils range from silty clay loams to clays. Deep Red Brown Terrace soil, Grey Piedmont soils and Acid Basin Clays are the major components of general soil types of the area. The general fertility including organic matter status is low. The soils are strongly acidic in reaction.

Location: Brahmanbaria district and minor area in Habiganj district.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.3 table p.27; Section 1.4 description p.34]

Section 1.5 — Land Levels in Relation to Flooding

The information has been provided in terms of depth of flooding phases. The terms used have the following meanings:

Highland (H): Land which is above normal flood-level.

Medium Highland (MH): Land which is normally flooded about 90 cm deep during the flood season.

Medium Lowland (ML): Land which is normally flooded between 90 cm and 180 cm deep during the flood season.

Lowland (L): Land which is normally flooded between 180 and 300 cm deep during the flood season.

Very Lowland (VL): Land which is normally flooded deeper than 300 cm during the flood season.

An additional class, Bottomland, is recognized for depression sites in any land level class which remains wet throughout the year. The Medium Highland land type is divided into two subclasses: MH-1, normally flooded up to about 30 cm deep; and MH-2, normally flooded between 30-90 cm deep.

Flood-levels in an area may vary by as much as a metre or more over different years, and may reach peak levels for only a few days at a time. These classes indicate the level of flooding which farmers expect when deciding which crops to grow in the kharif season on their different kinds of land, based on long experience of cultivation at particular sites.

Highland may be suitable for kharif or perennial dry land crops if the soils are permeable. Impermeable soils, or soils which can be made impermeable by puddling, may be suitable for transplanted Aus and/or Aman paddy if bunds are made to retain rainwater on fields.

Medium Highland is suitable for crops which can tolerate shallow flooding, such as broadcast or transplanted Aus paddy, jute and transplanted Aman paddy. Early kharif dry land crops which mature before flooding starts can be grown on permeable soils, and late kharif and early rabi dry land crops on soils which drain in September-October.

Medium Lowland is flooded too deeply for transplanted Aus or transplanted Aman paddy to be grown safely. Mixed broadcast Aus and deep-water Aman is a common practice; or long Aman seedlings may be transplanted if floodwater recedes early enough. Dry land rabi crops are widely grown on soils which drain in October or November.

Lowland is flooded too deeply for broadcast Aus or transplanted Aman to be grown. Deepwater Aman is typically grown on such land (though irrigated Boro paddy cultivation in the dry season now precludes deepwater Aman over considerable areas). Dry land rabi crops can only be grown if floodwater recedes before December.

Very Lowland is generally too deeply flooded for even deep-water Aman (due to early flooding, rapid flooding or wave action rather than depth alone, as in the Sylhet Basin). Where cultivated, it is generally used for irrigated Boro paddy, either HYV or local varieties.

Bottomland stays too wet for paddy to be sown broadcast. The traditional crop on such land is local Boro paddy, either not irrigated or irrigated by traditional low-lift irrigation devices. In a few areas where flooding does not exceed about 1.5 m deep, very long Aman paddy seedlings are transplanted early in the monsoon season.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.5, pp.37-38]

Section 1.6 — Soil Classification of Bangladesh

Results of reconnaissance soil surveys conducted in the recent past have enabled scientists to divide the country into 19 soil type units ("The Soils of East Pakistan", 1969: H. Brammer). These units, their characteristics, and places of occurrence are listed below. Source: Soil Resource Development Institute.

Soil Unit 1: Non-calcareous Alluvium.

Characteristics: Recent Tista, Brahmaputra and Jamuna alluvium. Mainly unstable char land.

Places of occurrence (District: Upazilas): Mymensingh: Nagarpur, Tangail, Kalihati, Gopalpur, Sarishabari, Madargonj, Melandah, Islampur and Dewangonj. Rangpur: Raumari, Nageswari, Bhurungamari, Kurigram, Ulipur, Chilmari, Gaibandah and Phulchari. Bogura: Saria Kandi and Dhunot. Pabna: Kazipur, Sirajgonj, Kamarkhonda, Belkuchi, Chauhali, Shahzadpur and Bera.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.38]

Soil Unit 2: Calcareous Alluvium.

Characteristics: Recent Ganges and lower Meghna alluvium. Part unstable char land. Part saline in the Meghna estuary.

Places of occurrence (District: Upazilas): Faridpur: Char Bhadrashan and Sadarpur. Noakhali: Sudharam, Hatiya, Ramgati and Sonagazi. Chattogram: Sandwip

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.38]

Soil Unit 3: Acid Sulphate Soils.

Characteristics: Sundarbans (mangrove swamp) soils with extremely high acidity (potential or actual) tidally flooded with blackish or saline water for part or all the year.

Places of occurrence (District: Upazilas): Khulna: Dacope and Sarankhola.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.38]

Soil Unit 4: Peat.

Characteristics: Permanently wet basin peat and muck, part with alluvial topsoil.

Places of occurrence (District: Upazilas): Faridpur: Mukusudpur, Gopalganj, Kotalipara, Kasiani, Rajoir and Madaripur. Sylhet: Daulatpur, Fultala, Terokheda, Dumuria and Mollahat. Bakerganj: Nazirpur. Jashore: Abhoynagar, Salika and Kalia.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.39]

Soil Unit 5: Grey Floodplain Soils.

Characteristics: Grey, finely mottled brown, seasonally flooded soils with seasonally acid top-soil and near neutral subsoils.

Places of occurrence (District: Upazilas): Dhaka: Keraniganj, Chapinawabgonj, Kaliganj, Dhamrai, Munshiganj, Gazaria, Manikganj, Saturia, Singair, Daulatpur, Gheor, Narayangonj, Futullah and Raipura. Mymensingh: Sarishabari, Madargonj, Tangail, Kalihati and Nagarpur. Cumilla: Homna, Chandpur, Matlabbazar and Bancharampur.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.39]

Soil Unit 6: Grey Floodplain Soils and Non-Calcareous brown floodplain soils.

Characteristics: Seasonally wet or shallowly flooded Grey Floodplain soils on lower ridges and in depressions with moderately well-drained, rather acid, brown loams on higher ridges.

Places of occurrence (District: Upazilas): Mymensingh: Kotwali, Islampur, Ishwargonj, Gouripur and Jamalpur. Dinajpur: Chirirbandar, Fulbari and Khansama. Rangpur: Nilphamari, Saidpur, Dimla, Domar, Jhaldaka, Kishoregonj, Gangachara, Mithapukur, Badarganj, Kaliganj, Hatibanda, Patgram, Palasbari and Sundargonj. Bogura: Sherpur.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.39]

Soil Unit 7: Mixed grey, Dark grey and Brown Floodplain soils.

Characteristics: Tista floodplain soils, Noncalcareous brown floodplain soils on higher ridges; seasonally wet or flooded mainly silty, mixed Grey Floodplain soils and Non-Calcareous Dark Grey Floodplain soils on lower ridges and in depressions.

Places of occurrence (District: Upazilas): Rangpur: Bhurungamari, Lalmonirhat and Fulbari.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.39]

Soil Unit 8: Grey Floodplain soils and Non-calcareous Dark grey floodplain soils.

Characteristics: Seasonally wet or shallowly flooded Grey Floodplain soils, mainly on ridges, and seasonally shallowly or deeply flooded Non-calcareous Dark Grey Floodplain Soils, mainly in basis.

Places of occurrence (District: Upazilas): Dhaka: Gazaria, Arai hazar and Narsingdi. Kishoreganj: Hossainpur, Mohanganj, Barhatta and Purbadhala. Mymensingh: Nandail, Phulpur, Jamalpur, Sherpur, Nakla and Melandah. Noakhali: Companyganj, Lakshmpur, Raipur, Begumganj, Senbag and Feni. Cumilla: Muradnagar, Burichang, Chauddagram, Laksam, Barura, Faridganj, Hajigonj and Nabinagar.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.39]

Soil Unit 9: Grey Floodplain Soils and Acid basin clays.

Characteristics: This occupies the eastern Surma- Kusiyara Floodplain in Sylhet and the Cumilla basin.

Places of occurrence (District: Upazilas): Kishoreganj: Bhairab Cumilla: Kotwali and Burichang. Sylhet: Balaganj, Biswanath; Fenchuganj, Zakiganj, Beanibazar, Habiganj, Chhatak, Jagannathpur, Moulvibazar and Rajanagar.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.40]

Soil Unit 10: Grey Piedmont Soils.

Characteristics: Grey mottled red or brown, strongly acid, loams to clays on seasonally wet or flooded piedmont plains adjoining the eastern hills.

Places of occurrence (District: Upazilas): Chattogram: Kotwali, Doublemoorings, Mirsarai, Sitakunda, Fatikchari, Raojan, Rangunia, Boalkhali, Patiya,Satkania, Chakaria, Kutubdia and Ramu. Chattogram Hill Tracts: Lama. Noakhali: Chhagalnaiya. Cumilla: Kasba and Chauddagram. Sylhet: Chunarughat, Maulvibazar, Kulaura, Kamalgonj and Barlekha.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.40]

Soil Unit 11: Acid basin clays.

Characteristics: Strongly acid heavy clays, part permanently wet.

Places of occurrence (District: Upazilas): Dhaka: Dohar and Serajdikhan. Kishoregonj: Astagram, Nikli, Itna,Khaliajuri, Madan, Kalmakanda and Durgapur. Cumilla: Nasirnagar. Sylhet: Nabiganj, Lakhai, Azmiriganj, Sunamganj, Tahirpur, Derai, Sulla and Jamalganj. Rajshahi: Naogaon, Atrai, Raninagar and Singra. Sirajgonj: Tarash.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.40]

Soil Unit 12: Non-Calcareous Dark grey Floodplain soils.

Characteristics: Dark grey, finely mottled brown, and brown soils with dark grey flood coatings, with seasonally acid top soils and near-neutral sub-soils. Mainly seasonally deeply flooded soils of the old Brahmaputra- Karatoya-Bangali (Part) and old Meghna estuarine floodplains.

Places of occurrence (District: Upazilas): Dhaka: Tejgaon, Lohajong, Tongibari, Araihazar, Rupaganj and Sonargaon. Kishoregonj: Kotwali, Pakundia, Karimgonj, Tarail, Bajitpur, Kuliarchar, Kathiadi, Netrakona, Atpara and Kendua. Mymensingh: Muktagacha, Fulbaria, Trisal, Sarisabari, Basail, Ghatail and Kalihati. Faridpur: Bhedarganj,Naria, Madripur and Palong. Noakhali: Ramganj. Cumilla: Chandina, Debidwar, Daudkandi, Kachua, Brahmanbaria, Sarail, Nasirnagar, Nabinagar and Bancharampur.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.40]

Soil Unit 13: Calcareous dark grey floodplain soils and calcareous brown floodplain soils.

Characteristics: Mainly dark grey or brown clays with dark grey flood coatings, some calcareous throughout some with seasonally acid top soils and calcareous substratum within 4 feet. Brown calcareous loamy soils on highest ridges and near river-banks.

Places of occurrence (District: Upazilas): Dhaka: Srinagar, Manikganj, Harirampur and Shivalaya. Mymensingh: Kotwali. Faridpur: Bhanga, Nagarkanda, Baliakandi, Muksudpur, Kotalipara, Kasiani, Kalkini, Goshairhat, Janjira and Shibchar. Rajshahi: Boalia, Paba, Puthia, Charghat, Bholahat, Shibganj, Gamustapur, Natore, Bagatipara, Lalpur, Baraigram and Gurudaspur. Pabna: Kotwali, Atgharia, Ishurdi, Chatmohar, Faridpur, Santhia, Bera and Sujanagar.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.41]

Soil Unit 14: Calcareous dark grey floodplain soils with lime kankar.

Characteristics: Mainly leached calcareous dark grey floodplain soils about half with a hard line kankar layer 2-6 feet deep.

Places of occurrence (District: Upazilas): Rajshahi: Bagmara, Durgapur.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.41]

Soil Unit 15: Non-calcareous brown floodplain soils and grey floodplain soils.

Characteristics: Brown soils are the dominant soils in the landscape and the grey soils are subordinate.

Places of occurrence (District: Upazilas): Mymensingh: Phulpur, Jamalpur, Madargonj and Dewanganj. Dinajpur: Birganj, Kaharole, Birol, Bochaganj, Thakurgaon, Baliadangi, Ranisankail, Haripur, Pirganj, Debiganj and Boda.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.41]

Soil Unit 16: Black Terai soils.

Characteristics: Seasonally wet, dark coloured rather acid, loamy soils on ridges, level areas and in depressions.

Places of occurrence (District: Upazilas): Dinajpur: Panc hagarh, Atwari and Tetulia.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.41]

Soil Unit 17: Brown Hill soils.

Characteristics: Brown, very strongly acid, mainly loamy soils.

Places of occurrence (District: Upazilas): Kishoreganj: Dinajpur. Mymensingh: Haluaghat, Sribordi and Nalitabari. Chattogram: Panchlaish, Hathazari, Mirsarai, Fatikchari, Rangunia, Anowara, Banskhali, Cox's Bazar, Ramu, Moheshkahli, Teknaf and Ukhiya, all thanas of Chattogram Hill Tracts. Noakhali: Chagalnaiya and Parsuram. Cumilla: Kotwali and Chauddagram. Sylhet: Kotwali, Gowainghat, Golapgonj, Jaintapur and Kanaighat.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.41]

Soil Unit 18: Red-brown Terrace soils.

Characteristics: Well to moderately well-drained, red and brown, strongly acid, clay loams and clays, part over compact Modhupur clay at 1-3 feet, pert over deeply mottled clay substratum.

Places of occurrence (District: Upazilas): Dhaka: Sripur, Kapasia, Kaligonj, Gazipur Sadar, Savar, Kaliakhair, Rupgonj, Shibpur and Monohardi. Mymensingh: Fulbaria, Trisal and Ghatail. Cumilla: Kasba. Rajshahi: Nowabgonj. Dinajpur: Chapinawabgonj, Gharaghat. Rangpur: Pirganj, Gobindaganj and Sadullapur.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.41]

Soil Unit 19: Grey Terrace soils.

Characteristics: Poorly drained, grey, mottled, acid silty soils over a grey, mottled clay substratum.

Places of occurrence (District: Upazilas): Dhaka: Gazipur Sadar. Kishoreganj: Durgapur. Mymensingh: Haluaghat, Sribordi and Nalitabari. Rajshahi: Tanore, Godagari, Nowabganj, Bholahat, Nachole, Shibganj, Porsha, Badalgachi, Niamatpur, Mohadevpur, Raninagar and Dhamoirhat. Dinajpur: Kotwali, Parbatipur, Phulbari and Hakimpur. Rangpur: Mithapukur, Gobindaganj, Sadullapur. Bogura: Panchbibi, Khetlal, Adamdigi, Dubchanchia, Kahalu, Kotwali, Shibganj and Nandigram. Sirajgonj: Tarash.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.6 table, p.42]

Section 1.7 — Crop Seasons and Seed Requirements

The country grows a wide variety of crops which are broadly classified, according to the season in which they are grown, into two groups. (a) Kharif crops: grown in the spring or summer season and harvested in late summer or early winter. (b) Rabi crops: sown in winter and harvested in the spring or early summer. Time of sowing/transplanting, time of harvest, and per-acre seed requirement for each crop are given in the Crop Calendar (Section 1.8) below.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.7, p.42]

Section 1.8 — Crop Calendar of Bangladesh (86 crops)

For each crop: time of sowing/transplanting, time of harvest, and per-acre seed requirement. Where a crop has multiple varieties or planting windows (e.g. kharif/rabi, local/HYV/hybrid), each is listed separately.

1. Aus paddy

(a) Local Broadcast Sowing/Transplanting: Mid-March to Mid-April | Harvest: Mid July to Early August | Seed requirement: 28-37 kg in case of broadcasting: 14-19 kg in case of line sowing

(b) HYV Transplant Sowing/Transplanting: Mid-March to Mid-April | Harvest: July to August | Seed requirement: 11-14 kg

(c) HYV Broadcast Sowing/Transplanting: Mid-March to Mid-April | Harvest: Late July to August | Seed requirement: 8-9 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.42]

2. Aman paddy

(a) Local Transplant Sowing/Transplanting: End June to Early September | Harvest: December to early January | Seed requirement: 9-11 kg

(b) Local Broadcast Sowing/Transplanting: Mid-March to Mid-April | Harvest: Mid November to Mid-December | Seed requirement: 33-37 kg

(c) HYV Transplant Sowing/Transplanting: Late June to Mid-August | Harvest: December to early January | Seed requirement: 8-9 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.42]

3. Boro Paddy

(a) Local Sowing/Transplanting: Mid November to Mid-January | Harvest: April to May | Seed requirement: 8-11 kg

(b) HYV Sowing/Transplanting: December to Mid-February | Harvest: Mid-April to June | Seed requirement: 8-11 kg

(c) Hybrid Sowing/Transplanting: December to Mid-February | Harvest: Mid-April to June | Seed requirement: 8-11 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

4. Wheat

Sowing/Transplanting: November to December | Harvest: March to Mid-April | Seed requirement: (a) 28-33 kg if sown without irrigation. (b) 33-37 kg if sown with 1-2 times irrigation. (c) 37-47 kg if sown with 4-5times irrigation.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

5. Maize (Rabi)

Sowing/Transplanting: Mid October to Late December | Harvest: Early April to End May | Seed requirement: 7-9 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

6. Jowar

Sowing/Transplanting: Mid-April to June | Harvest: Mid-August to Mid-October | Seed requirement: 7-9 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

7. Kaon

Sowing/Transplanting: (a) November to December, in case of low land. Mid-March to Mid-May, in case of hilly land | Harvest: Mid-March to Mid-June Mid-June to Mid-August | Seed requirement: 5-6 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

8. Cheena

Sowing/Transplanting: November to Mid-December | Harvest: Mid-February to Mid-April | Seed requirement: 5-6 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

9. Barley

Sowing/Transplanting: Mid-October to Mid-December | Harvest: Mid-February to Mid-April | Seed requirement: 23-28 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

10. Potato

Sowing/Transplanting: Mid-September to November | Harvest: Mid-January to March | Seed requirement: 4-6 Ql tuber

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

11. Sweet Potato

Sowing/Transplanting: Mid-September to Mid- December | Harvest: Mid-February to Mid-April | Seed requirement: 1200-1400 cuttings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

12. Jute

(a) White (Capsularis) Sowing/Transplanting: Early March to Mid-April | Harvest: July to August | Seed requirement: 4-5 kg

(b) Tossa (Olitorius) Sowing/Transplanting: Mid-April to Early May | Harvest: August to September | Seed requirement: 4-5 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

13. Sun hemp

(a) Kharif Sowing/Transplanting: Mid-March to Mid-June | Harvest: Mid July to Mid-October | Seed requirement: 3-4 kg

(b) Rabi Sowing/Transplanting: Mid-September to Mid- November | Harvest: Mid-February to Mid-April | Seed requirement: 3-4 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

14. Cotton

(a) Kharif Sowing/Transplanting: Mid-April to Mid-June | Harvest: Mid-October to Mid-December | Seed requirement: 4-5 kg

(b) Rabi Sowing/Transplanting: Mid-August to Mid-October | Harvest: Mid-February to Mid-April | Seed requirement: 7-8 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

15. Rape Seed and Mustard

Sowing/Transplanting: Mid October to Mid-November | Harvest: Late January to Mid-February | Seed requirement: 3-4 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

16. Groundnut

(a) Kharif Sowing/Transplanting: Mid-June to Mid-July | Harvest: Late October to Mid-November | Seed requirement: 37 kg

(b) Rabi Sowing/Transplanting: Mid November to Late December | Harvest: Late April to Mid-June | Seed requirement: 37 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

17. Teel

(a) Kharif Sowing/Transplanting: Early February to Mid-March | Harvest: Early May to Mid-June | Seed requirement: 3-4 kg

(b)Rabi Sowing/Transplanting: Beginning September to Late October | Harvest: December | Seed requirement: 3-4 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

18. Linseed

Sowing/Transplanting: Mid October to Mid-December | Harvest: Mid-March to Mid-May | Seed requirement: 4-5 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.43]

19. Sunflower

Sowing/Transplanting: Mid November to Mid December | Harvest: Late February to Mid March | Seed requirement: 7 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

20. Arhar (Pigeon pea)

Sowing/Transplanting: Mid April to Mid June | Harvest: Mid January to Mid March | Seed requirement: 6-8 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

21. Mashkalai (Black Gram)

Sowing/Transplanting: Mid August to Mid September | Harvest: Last three week of November | Seed requirement: 8-9 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

22. Soyabean

(a) Kharif Sowing/Transplanting: Beginning June to Late July | Harvest: Late September to Mid November | Seed requirement: 10 kg

(b) Rabi Sowing/Transplanting: Last week of December | Harvest: Mid April to Early May | Seed requirement: 10 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

23. Barbati

Sowing/Transplanting: (a) Mid September to Mid January | Harvest: Mid December to Mid April | Seed requirement: 11-14 kg

(cont.) Sowing/Transplanting: (b) Mid January to Mid June | Harvest: - | Seed requirement: -

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

24. Mung bean

Sowing/Transplanting: Mid August to Mid September | Harvest: Mid November to Mid December | Seed requirement: 8-10 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

25. Masur (Lentil)

Sowing/Transplanting: Mid October to Mid November | Harvest: Early February to Early March | Seed requirement: 11-14 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

26. Kheshari

Sowing/Transplanting: Mid October to Mid December | Harvest: Mid February to Mid April | Seed requirement: 9-11 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

27. Chickpea (Chola)

Sowing/Transplanting: Mid October to Late November | Harvest: Early March to Early April | Seed requirement: 14-17 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

28. Motor (Field Pea)

Sowing/Transplanting: Mid October to Late November | Harvest: Early March to Early April | Seed requirement: 9-14 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

29. Ginger

Sowing/Transplanting: Mid March to Mid May | Harvest: Mid December to Mid March | Seed requirement: 8-11 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

30. Turmeric

Sowing/Transplanting: Mid April to Mid June | Harvest: Mid December to Mid March | Seed requirement: 8-11 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

31. Onion

Sowing/Transplanting: Beginning October to Early December | Harvest: Late April to Mid June | Seed requirement: 2-3 kg in case of seeds 2-3 Ql. in case of bulb

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

32. Garlic

Sowing/Transplanting: Mid October to Mid December | Harvest: Mid February to Mid March | Seed requirement: 0.37 Ql.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

33. Coriander seed

Sowing/Transplanting: October to December | Harvest: Mid February to Mid March | Seed requirement: 6-8 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

34. Chilies:

(a) Kharif Sowing/Transplanting: Mid April to Mid July | Harvest: 3-4 months after sowing | Seed requirement: 1 kg in case of seeds

(b) Rabi Sowing/Transplanting: Mid November to Mid January | Harvest: 3-4 months after sowing | Seed requirement: Seedlings raised from approximately 100 grams seeds

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

35. Sugarcane

Sowing/Transplanting: Early Mid October to Mid December | Harvest: Mid October to Mid April | Seed requirement: 19 Ql cuttings

(cont.) Sowing/Transplanting: Late Mid January to Mid March | Harvest: Mid October to Mid April | Seed requirement: 22 Ql cuttings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

36. Tobacco

Sowing/Transplanting: Mid October to Mid December | Harvest: Mid February to Mid April | Seed requirement: 23-35 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.44]

37. Cauliflower

Sowing/Transplanting: Late October to Mid November | Harvest: Early January to Early March | Seed requirement: (a) Seeds: 105-117 gram (b) Seedlings: 8-10 thousand

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

38. Cabbage

Sowing/Transplanting: Late October to Mid November | Harvest: Early January to Early March | Seed requirement: (a) Seeds: 105-117 gram (b) Seedlings: 6-8 thousand

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

39.Oal Kapi (Knol Khol)

Sowing/Transplanting: Mid September to third week of December | Harvest: December to Mid February | Seed requirement: (a) Seeds: 220-350 grams (b) Seedlings: 25-30 thousand

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

40. Chinese Cabbage

Sowing/Transplanting: Mid September to Early November | Harvest: Late January to Mid March | Seed requirement: Seeds: 150 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

41. Tomato

Sowing/Transplanting: Mid August to Late November | Harvest: Beginning December to Mid January | Seed requirement: (a) Seeds: 50-70 grams (b) Seedlings: 6-8 thousand

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

42. Radish :

Sowing/Transplanting: Mid August to Early October | Harvest: Early January to Mid February | Seed requirement: 3-4 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

43. Carrot :

Sowing/Transplanting: September to Mid December | Harvest: December to January | Seed requirement: 1050 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

44. Shalgam : (Turnip)

Sowing/Transplanting: September to Mid December | Harvest: Mid November to Mid February | Seed requirement: 525 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

45. Beet :

Sowing/Transplanting: Mid September to Mid December | Harvest: Mid December to Mid April | Seed requirement: 3-4 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

46. Lettuce :

Sowing/Transplanting: Mid October to Mid December | Harvest: Mid December to Mid February | Seed requirement: (a) Seeds: 70 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

47. Brinjal :

Sowing/Transplanting: October to Mid November | Harvest: Late November to Mid April | Seed requirement: (a)Seeds: 75 grams (b) Seedlings: 6-8 thousand

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

48. Palongsak : (Spinach)

Sowing/Transplanting: September to Mid November | Harvest: 1-4 months after sowing | Seed requirement: 9 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

49. Lalsak (Red spinach) / Kangkong (Kachusak, water spinach)

Lalsak Sowing/Transplanting: Throughout the year | Harvest: Throughout the year | Seed requirement: 400 grams

Kangkong (Kachusak) Sowing/Transplanting: Mid March to Early August | Harvest: EarlySeptember to Early November | Seed requirement: not available

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

50. Green peas :

Sowing/Transplanting: Late half of November | Harvest: Late February to Mid April | Seed requirement: 14-15 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

51. Water gourd :

Sowing/Transplanting: Mid July to Early November | Harvest: Early January to Early March | Seed requirement: 250 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

52. Uchcheya : (Bitter gourd)

Sowing/Transplanting: Mid February to Early June | Harvest: Mid June to Mid October | Seed requirement: 800 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

53. Kakor :

Sowing/Transplanting: Mid December to Mid March | Harvest: May to July | Seed requirement: 50 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

54. Beans :

Sowing/Transplanting: Late June to Early September | Harvest: Late November to Mid April | Seed requirement: 800 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.45]

55. Cucumber :

Sowing/Transplanting: Mid September to Mid October | Harvest: Mid January to Mid June | Seed requirement: 150 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

56. Lady's finger :

Sowing/Transplanting: Mid April to Mid June | Harvest: June to Mid September | Seed requirement: 2-3 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

57. Patal :

Sowing/Transplanting: Mid August to Mid October | Harvest: Mid January to March | Seed requirement: 1-2 thousand cuttings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

58. Karala :

Sowing/Transplanting: Mid April to Mid June | Harvest: Mid June to Mid August | Seed requirement: 800 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

59. Chichinga : (Snake Gourd)

Sowing/Transplanting: Mid February to End April | Harvest: July to September | Seed requirement: 800 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

60. Kakrol :

Sowing/Transplanting: Mid April to Mid June | Harvest: Mid July to Mid September | Seed requirement: 400 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

61. Yard long bean:

Sowing/Transplanting: Mid October to End November | Harvest: Early January to Mid April | Seed requirement: 4 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

62. Jhinga : (Ribbed Gourd)

Sowing/Transplanting: Mid April to Mid July | Harvest: Mid June to Mid August | Seed requirement: 1-2 kg

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

63. Puisak : (Indian spinach)

Sowing/Transplanting: Mid March to Early June | Harvest: Late August to Mid November | Seed requirement: 200 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

64. Chal Kumra : (White Gourd)

Sowing/Transplanting: Mid March to Mid June | Harvest: Mid July to Mid October | Seed requirement: 90 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

65. Sweet Gourd:

Sowing/Transplanting: Second half of November | Harvest: EarlyApril to Mid June | Seed requirement: 200 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

66. Mukhi Kachu : (Taro)

Sowing/Transplanting: Mid April to Mid June | Harvest: Mid September to Mid February | Seed requirement: 2-3 Ql

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

67. Danta : (Amaranth)

Sowing/Transplanting: Mid February to End June | Harvest: Late August to Mid November | Seed requirement: 350 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

68. Water Melon :

Sowing/Transplanting: Mid October to Mid January | Harvest: Mid February to Mid June | Seed requirement: 350 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

69. Melon :

Sowing/Transplanting: Mid October to Mid January | Harvest: Mid February to Mid June | Seed requirement: 350 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

70. Kharmuj : (Musk Melon)

Sowing/Transplanting: Mid October to Mid January | Harvest: Mid February to Mid June | Seed requirement: 350 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

71. Pineapple :

Sowing/Transplanting: a) Mid March to Mid May | Harvest: May to July | Seed requirement: 10-14 thousand saplings

(cont.) Sowing/Transplanting: b) Mid September to Mid November | Harvest: May to July | Seed requirement: 10-14 thousand saplings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

72. Banana :

Sowing/Transplanting: a) Mid January to Mid March | Harvest: December to February | Seed requirement: 700-1700 suckers

(cont.) Sowing/Transplanting: b) Mid September to Mid November | Harvest: December to February | Seed requirement: 700-1700 suckers

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

73. Papaya :

Sowing/Transplanting: Mid June to End August | Harvest: Early July to End August | Seed requirement: 150 grams

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

74. Litchi :

Sowing/Transplanting: Mid April to Mid June. | Harvest: Mid April to Mid June | Seed requirement: 48 Saplings

(cont.) Sowing/Transplanting: (b) Mid September to Mid November | Harvest: Mid April to Mid June | Seed requirement: 28 Saplings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

75. Mango :

Sowing/Transplanting: a) Mid April to Mid June | Harvest: Mid April to Mid June | Seed requirement: 28 Saplings

(cont.) Sowing/Transplanting: b) Mid September to Mid November | Harvest: Mid April to Mid June | Seed requirement: -

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

76. Jackfruit :

Sowing/Transplanting: a) Mid April to Mid June | Harvest: Mid April to Mid July | Seed requirement: 28 Saplings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.46]

77. Safeda (Sapota)

Sowing/Transplanting: Mid-April to Mid-July | Harvest: (a) Mid-January to Mid-March | Seed requirement: 48 Saplings

(cont.) Sowing/Transplanting: - | Harvest: (b) Mid-May to Mid-July | Seed requirement: -

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.47]

78. Pamelo

(a) Sowing/Transplanting: Mid-April to Mid-July | Harvest: Mid-May to Mid-June | Seed requirement: 108 Saplings

(b) Sowing/Transplanting: Mid-September to Mid-November | Harvest: Mid-May to Mid-June | Seed requirement: -

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.47]

79. Guava

Sowing/Transplanting: Mid-April to Mid-July | Harvest: Throughout the year; peak season June to September | Seed requirement: 134 Saplings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.47]

80. Lime and Lemon

Sowing/Transplanting: Mid-April to Mid-July | Harvest: Throughout the year; peak season June to September | Seed requirement: 302 Saplings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.47]

81. Date Palm

Sowing/Transplanting: Mid-April to Mid-July | Harvest: Mid-February to Mid-March | Seed requirement: 64-78 Seedlings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.47]

82. Dalim (Pomegranate)

(a) Sowing/Transplanting: Mid-April to Mid-July | Harvest: Throughout the year | Seed requirement: 193 Sapling

(b) Sowing/Transplanting: Mid-September to Mid-November | Harvest: - | Seed requirement: -

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.47]

83. Ber

(a) Sowing/Transplanting: Mid-April to Mid-July | Harvest: February to March | Seed requirement: 69 Sapling

(b) Sowing/Transplanting: Mid-September to Mid-November | Harvest: - | Seed requirement: -

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.47]

84. Sharifa (Custard Apple)

Sowing/Transplanting: Mid-September to Mid-November | Harvest: After 7 years fruiting round the year | Seed requirement: 64-78 Seedlings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.47]

85. Coconut

Sowing/Transplanting: May to Mid-July | Harvest: After 7 years fruiting round the year (-do-) | Seed requirement: 64-74 Seedlings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.47]

86. Betelnut

Sowing/Transplanting: Mid-October to Mid-December | Harvest: Mid-October to Mid-December | Seed requirement: 1200-1400 Seedlings

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.8 table, p.47]

Section 1.9 — Physiography of Bangladesh

Bangladesh forms the largest delta in the world and is situated between 88 degrees 10' and 92 degrees 41' East longitude and between 20 degrees 34' and 26 degrees 38' North latitude. The great delta is flat throughout and stretches from near the foot-hills of the Himalayan Mountains in the north to the Bay of Bengal in the south.

The vast plain is washed by mighty rivers - the Meghna, the Padma, the Jamuna and the Karnafuli and their numerous tributaries. Tropical monsoon rains drench the land and the rivers; onrush of rain waters in summer overflows their banks, flooding low and outlying areas every year.

The monotony of flatness is relieved inland by two elevated tracts - the Modhupur and the Barind tracts - and, on the north-east and south-east, by rows of hilly forests. The great plain lies almost at sea level along the southern coast and rises gradually towards the north. The maximum elevation above mean sea level is 4034 feet at Keocradang Hill in Rangamati Hill district. Topography can be divided into five classes:

1. High Land: Relatively high, cannot hold water during monsoon (some retained by raising "bandhs" around fields). Spreads over Modhupur Garh (Tangail, Mymensingh), Bhaoal's Garh (Gazipur, Dhaka), Barind tract (Rajshahi Division), Lalmai area (Cumilla), and "Tilla" areas (Sylhet, Moulvi Bazar, Habiganj).

2. Medium Highland: Normally flooded up to about 90 cm depth during the rainy season for more than two weeks continuously. Spreads over Barishal division, major parts of Khulna division, northern Rajshahi division, and parts of Gazipur, Narsindi, Noakhali, Feni, Lakshmipur, Cumilla and Habiganj district.

3. Medium Lowland and Low Land: Medium Lowland is normally flooded between 90cm and 180cm depth; Low Land between 180cm and 275cm depth during the monsoon. Spreads over major parts of Cumilla, Brahmanbaria, Chandpur, Gopalgonj district and parts of Lakshmipur, Noakhali, Serajganj, Natore, Naogaon, northern Khulna and Bagherhat, and minor parts of Jashore, Kishorganj and Habiganj district.

4. Very Low Land: Haors, bils, canals and other low-lying areas resembling large lakes during the rainy season; water depth may rise as high as 30 feet. Most haors and bils lie in Sylhet division and in Kishoreganj and Netrokona district.

5. Hilly Land: Spreads over Rangamati, Bandarban and Khagrachari Hill Districts, parts of Chattogram, northern Mymensingh, north and south Sylhet division, eastern border of Cumilla and north-eastern strip of Feni district.

[Source: Yearbook of Agricultural Statistics 2020, Bangladesh Bureau of Statistics (BBS), 32nd Series, Section 1.9, p.49]